

Configuration

Configure the neural network and the image source.

Select Model Weight

deepglobe_finetuned_model.pth

Image Source

- ☐ Fetch Random Image (Matches Model)
- ☒ Upload Custom Image

Upload Image

 3594_sat.jpg
383.5KB

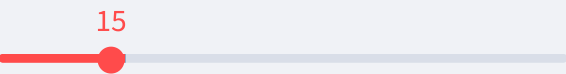
+

Resilience Settings

Attack Simulation Type

- ☒ Targeted (Choke Points)
- ☐ Random (Chance)

Nodes to Remove (%) [Random Attack Only]



Geospatial Network Resilience Dashboard

Analyze satellite imagery, extract topology-preserving road networks, and simulate cascading infrastructure failures.

Start Analysis Pipeline

> Terminal Execution Logs

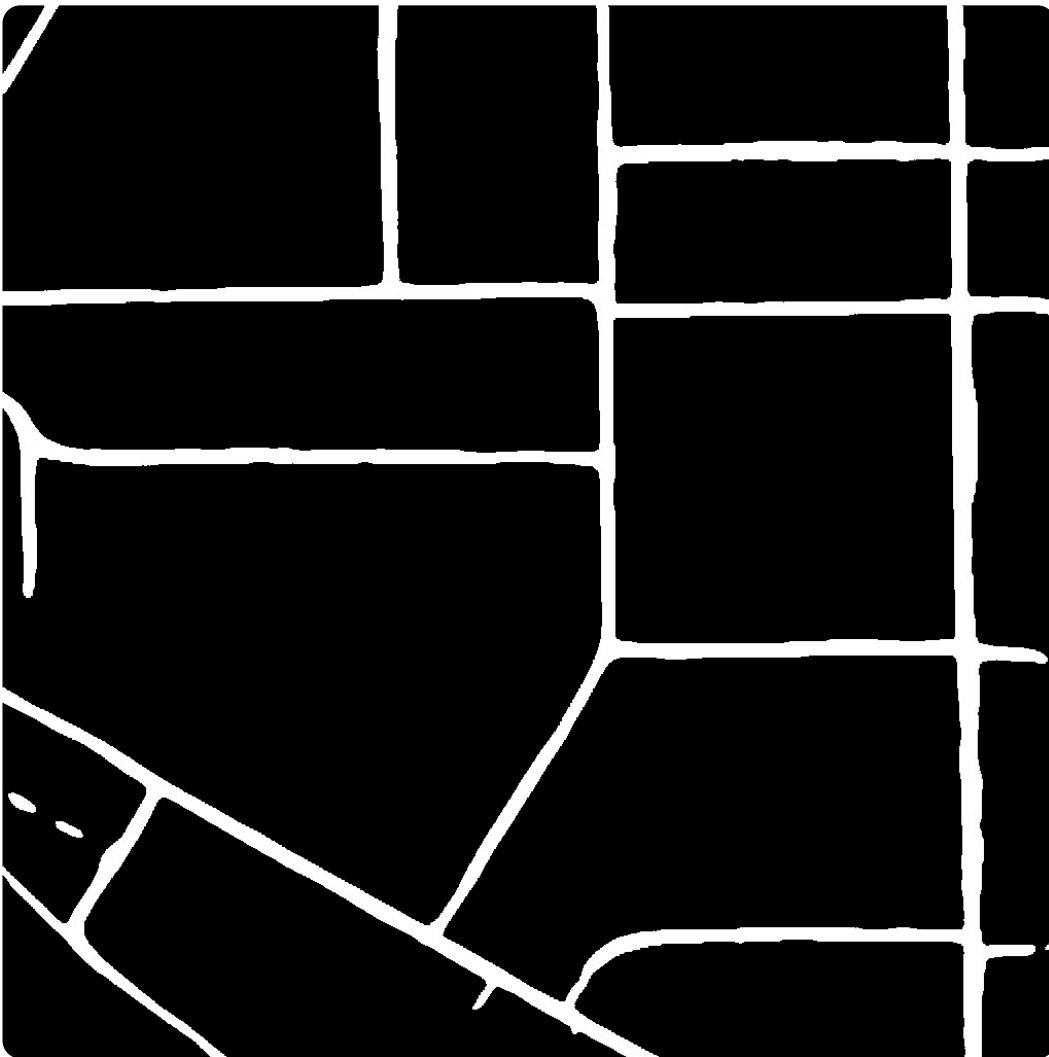
Pipeline Execution Complete!

Step-by-Step Visualizations

1. Input Satellite Image



2. AI Extracted Mask (UNet++)



3. Graph Analysis & Vulnerabilities



4. Threats & Results Summary

Simulation: Targeted (Choke Points)

Total Network Capacity Lost

50.0%

Cascading Steps

1

Roads Destroyed

6

Interactive 3D Network Map

Explore the extracted graph nodes and choke points projected onto Earth coordinates.

